

2024



EXISTS

Building an AI Solution for Quick Game Creation with Natural Language Using AWS with Exists

Learn how Exists in the gaming industry is building an AI-powered solution on AWS that can use natural language text prompts to create 3D games in minutes.

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5
minutes or less to create a 3D video game



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4,500
users in the test community with hundreds of active users



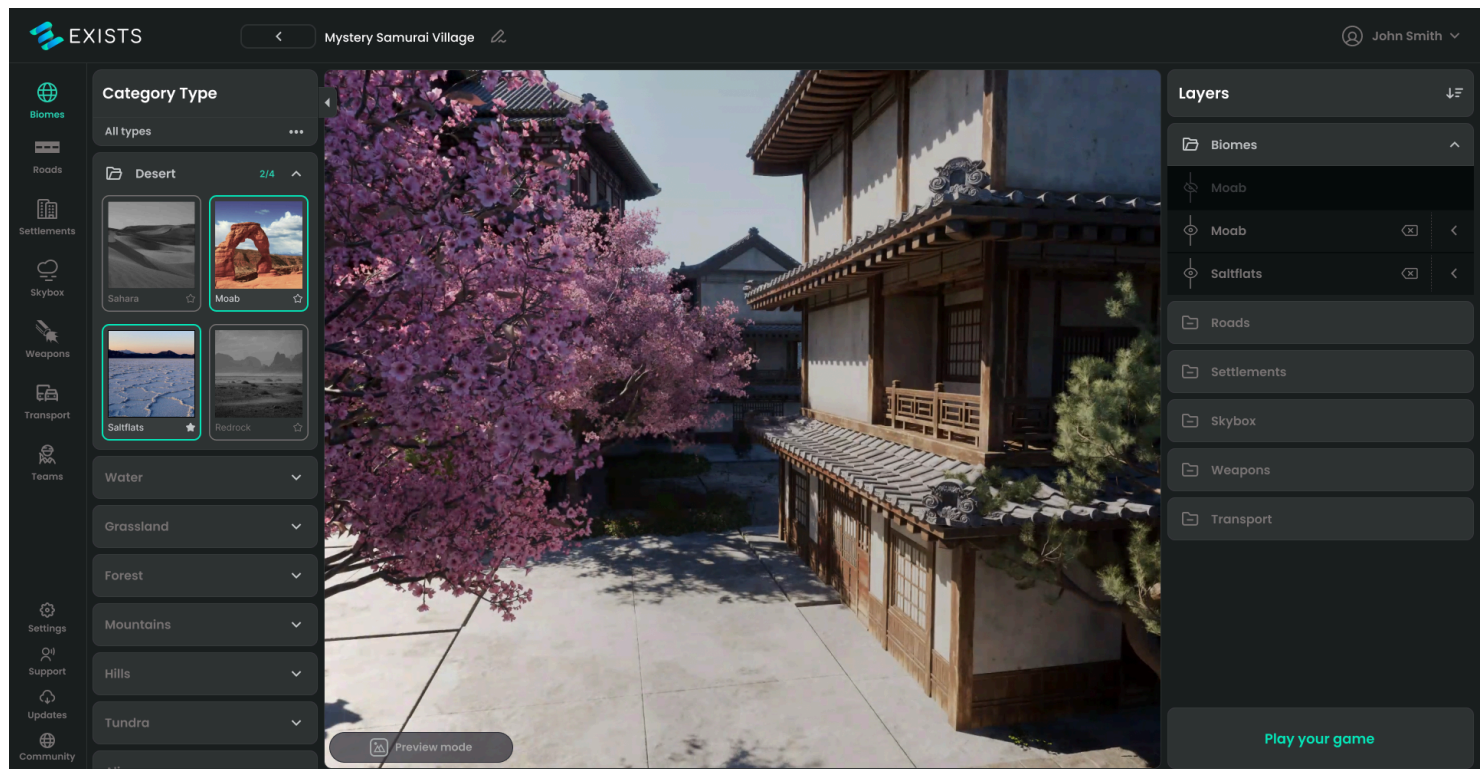
Need more info? Highlight any text to get an explanation generated with AWS generative AI.



Overview

Creating high-quality, complex 3D games requires a significant investment of time, expertise, and resources. [Exists](#) wants to ease the creative path for both seasoned game developers and aspiring creators who have great ideas but little technical experience. Exists has built a generative artificial intelligence (AI) text-to-game platform on Amazon Web Services (AWS) that will help users create fully playable multiplayer games, such as team deathmatch or battle royale, by simply entering text prompts. The company hopes to transform the game creation world with this solution by making advanced game development capabilities accessible to all.

“Game creation is a complicated process that requires heavy technical knowledge, good artistic capabilities, and huge teams,” says Dr. Yotam Hechtlinger, CEO and founder of Exists. “We want to make it possible for people with zero game creation experience to implement their creative visions as easily as logging into our website.”



Opportunity | Using AWS Infrastructure to Create a Generative AI Solution for Exists

Exists has combined novel AI research with a cloud-backed interface to build a solution that will help users create high-end interactive experiences. Since March 2023, the company has been creating its unique text-to-game platform, a potential accelerator in the gaming industry.

To achieve this, Exists has developed a neural architecture that combines multiple machine learning (ML) models into a game creation pipeline. At its core is a proprietary AI technology that integrates ML directly with game engines. As an added enhancement, Exists leverages recent advances in generative AI—such as diffusion models and large language models (LLMs)—to increase flexibility and offer creators even more versatility within the platform.

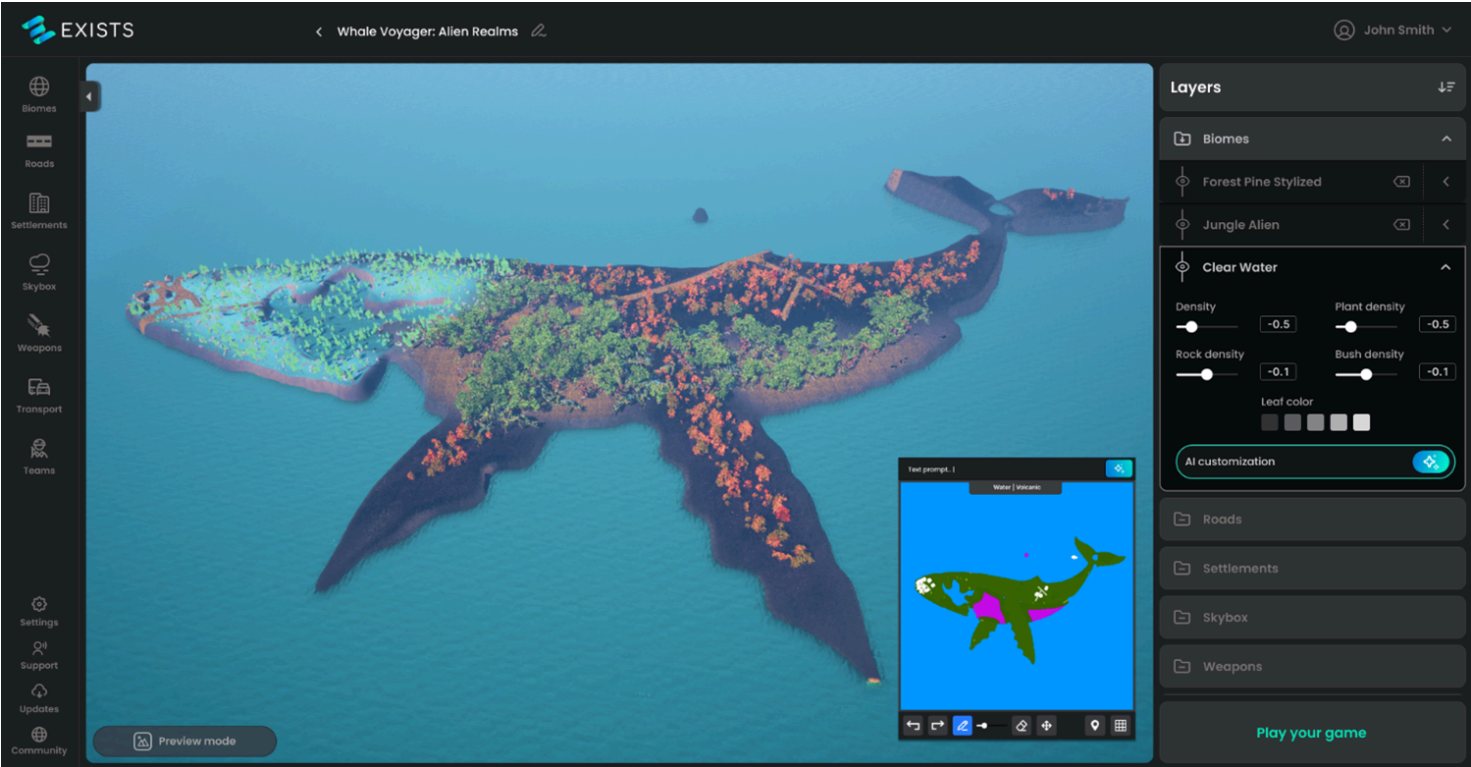
To support these advanced components, Exists relies on established generative AI infrastructure on AWS. For components using LLMs, Exists uses [Amazon Bedrock](#)—a fully managed service that offers a collection of pre-trained foundation models—to efficiently compare and test the latest models, finding its best fit. For official releases, Exists uses [Amazon SageMaker](#)—a service to build, train, and deploy machine learning models.



deploy ML models—offers Exists a fully managed ML infrastructure at scale, facilitating production-ready deployment without the need for intensive manual management. “This infrastructure allows us to deploy models in a production environment seamlessly,” says Sivane Marciano, ML engineer at Exists. “It’s also more cost-effective than independently training models.”

Solution | Creating a 3D Video Game in Less Than 5 Minutes Using Exists Text-to-Game Platform

Game creators—whether newbies or experts—can create a game with Exists by entering text prompts. Exists’ generative AI solution transforms these prompts into playable elements and complex environments. The creator, working through a web interface, can then edit the game, compile it, and distribute it to gamers worldwide.



Exists develops the solution using its ML and generative AI pipeline running on [Amazon Elastic Compute Cloud](#) (Amazon EC2), which provides secure and resizable compute capacity for virtually any workload. The trained models and other data are stored on [Amazon Simple Storage Service](#) (Amazon S3), object storage built for any amount of data from anywhere.



The platform is hosted entirely on the cloud and is accessible to the users globally. Exists also uses [Amazon Route 53](#)—a reliable and cost-effective way to route end users to internet applications—to route game creators across regions to its solution. This setup allows consumers to experience high-quality game development without needing high-end hardware.

The games’ scalable distribution is made possible by [Amazon GameLift](#), which offers game servers toughened to handle multiplayer games at a massive scale while lowering the cost per player. Exists can thus deploy and distribute multiplayer games worldwide for a scalable and seamless gaming experience.

The creation process is managed on the cloud because developing high-quality, interactive games requires substantial computational resources. For instance, the backend technology needed to automate this process involves multiple servers and terabytes of storage that must be globally readily available upon request. Scalable cloud infrastructure on AWS makes it possible for Exists to provide these resources for user-generated game creation. “This infrastructure was simply not possible a few years ago, and making the solution simple for consumers is challenging in terms of DevOps,” says Hechtlinger. “The joint effort of Exists technology and AWS infrastructure makes this solution available at scale to end users.”

A game creator can generate a fully working multiplayer game in less than 5 minutes. This democratizes the game creation process and makes it easier for new game creators—who might not have technical expertise or access to expensive, high-end hardware—to bring about their gaming ideas. Beyond individual creators, Exists is working with major gaming studios to integrate its technology as a user-generated content platform within existing games, enabling franchise communities to create and modify content in new ways. Expert game creators can also use the solution to rapidly experiment with and validate ideas, assessing potential and building proof-of-concept versions that can evolve over time. “It is a paradigm change in game creation,” says Hechtlinger. “We’ve made the barrier so low that it will unlock a game creation experience that wasn’t accessible before.”

Outcome | Scaling for Open Beta Testing

The company started closed beta testing in September 2024. The beta solution has 4,500 users in its test community and hundreds of active users who can bring their gaming concepts to life.

Exists continues to scale its cloud infrastructure to prepare for an open beta launch. It is looking to handle larger traffic volumes and is working on optimizing its use of AWS resources. “I’m excited to assist gamers in getting involved in game design itself because there is something pure and entertaining about the process of game design,” says Hechtlinger. “I hope the game creation experience itself will be changed 5 years from now.”

About Exists

Israel-based Exists builds applied artificial intelligence (AI) to change how games are created. It combines cutting-edge AI research with an intuitive cloud-based interface to help users easily create and edit high-end interactive experiences.

AWS Services Used



Amazon GameLift

Amazon GameLift empowers you to build and scale the world's most demanding games. With predictive game server scaling, enterprise-grade security, and matchmaking for the largest crowds, Amazon GameLift takes the uncertainty out of launch day.

[Learn more »](#)

Amazon Bedrock

Amazon Bedrock helps you rapidly adapt and take advantage of the latest generative AI innovations with easy access to a choice of high-performing FMs from leading AI companies like AI21 Labs, Anthropic, Cohere, Meta, Mistral AI, Stability AI, and Amazon.

[Learn more »](#)

Amazon SageMaker

Amazon SageMaker is built on Amazon's two decades of experience developing real-world ML applications, including product recommendations, personalization, intelligent shopping, robotics, and voice-assisted devices.

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Amazon Route 53

Amazon Route 53 provides highly available and scalable Domain Name System (DNS), domain name registration, and health-checking cloud services.

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